Forecasting Stock Prices Using ARIMA and Benchmark Methods

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1 Introduction

Artificial Intelligence (AI) has been a hot topic in recent years, with its wide applications in scientific discovery, advanced cybersecurity, and personalized AI agents. In the stock market, technology stocks have accounted for 32% of global equity returns and 40% of total U.S. equity market returns since 2010 (Goldman Sachs, 2024), playing a dominant role in modern financial markets. Large-cap firms, such as Apple and Microsoft, represent a substantial share of total market capitalization; consequently, their performance increasingly shapes the trajectory of major stock indices, including the S&P 500 and the NASDAQ Composite (Yahoo Finance).

A technological breakthrough is broadly defined as a significant and sudden advancement in science or engineering that introduces new capabilities, reshapes industries, and overcomes previous technical limitations (Sandle, 2025). The launch of ChatGPT was one of the first moments when the general public experienced the transformative potential of artificial intelligence in everyday life on a large scale. Therefore, in this report we consider the release of ChatGPT to represent a major technological breakthrough.

The adoption of AI applications has reshaped market narratives surrounding technological growth and productivity (Amin et al., 2024). Given that technology companies are largely valued based on expectations of future innovation and scalability, breakthroughs can trigger revisions in investor sentiment, leading to movement in stock prices.

With the crucial role of technology stocks in the broader financial market and the rapid emergence of AI products, this report investigates whether technological breakthroughs affect the technology-sector stock market, specifically on the Technology Select Sector SPDR Fund (XLK) from 2017 to 2026. The main research questions are:

- Did the release of ChatGPT altered the forecastability of the XLK stock prices or returns?
- How well do ARIMA models forecast XLK compared with simple benchmark methods?

Specifically, this report examines

1) Whether pre-AI stock price data can effectively forecast post-AI XLK price behavior.
2) Which model best predicts XLK price dynamics.

2 Background

Financial time series forecasting has long been important in quantitative finance because investors rely on forecasts for trading and portfolio decisions. Since stock prices are sequential and time-dependent, many studies model stock prices as time series processes to predict future movements. Among traditional forecasting approaches, the Autoregressive Integrated Moving Average (ARIMA) model is one of the most widely used methods because it can transform non-stationary financial data into stationary series and capture autocorrelation patterns.

Previous studies have evaluated the effectiveness of ARIMA models in stock market prediction. Mondal et al. (2014) examined ARIMA forecasting performance across 56 Indian stocks from multiple sectors and found that ARIMA achieved relatively strong short-term forecasting accuracy, especially in the Information Technology sector. Their study also emphasized the importance of model selection using AICc criteria. However, prior research has also suggested that simple benchmark methods may perform similarly to more complex time series models. Kulkarni et al. (2020) compared ARIMA, Simple Moving Average (SMA), and Holt-Winters models and found that SMA produced the lowest prediction error for short-term stock forecasting. These findings suggest that stock prices may behave similarly to random walks, making it difficult for ARIMA models to substantially outperform simpler forecasting methods.

Previous studies have also suggested that major technological innovations can reshape stock market valuation and investor expectations. Laitner and Stolyarov (2003) argue that technological revolutions may lead to market revaluation by increasing expectations for future productivity and altering sector-wide valuation dynamics. This framework provides theoretical support for investigating whether generative AI corresponds to a detectable shift in XLK price dynamics following the release of ChatGPT.

The emergence of generative AI has intensified market attention toward technology firms and AI-related infrastructure companies. The launch of ChatGPT in November 2022 significantly increased OpenAI's valuation and accelerated investment in AI technologies. At the same time, firms associated with AI infrastructure, particularly NVIDIA, also experienced substantial market revaluations(Morningstar, 2026).

Understanding whether such a technological breakthrough corresponds to a lasting change in market behavior is crucial for investors and policymakers. Persistent shifts in return patterns can affect risk assessment, portfolio allocation, and forecasting models in financial market analysis. Financial return series are time-dependent and often exhibit changes in volatility and growth dynamics. Applying time series forecasting methods allows us to evaluate whether the emergence of generative AI changed the predictability of technology-sector stock prices, provides a further insights for the shareholders.

3 Data

3.1 Data Overview

This report uses the daily XLK fund return price from Investing.com, a financial data platform that offers real-time data. The XLK stock price is collected by S&P Dow Jones Indices and State Street Global Advisory. S&P Dow Jones Indices is in charge of creating the company benchmark on entering the technology company stock pool. The State Street Global Advisors selected particular stocks into the combination of the XLK stock.

We chose the XLK because it's a comprehensive fund that captures an effective representation of the technology sector of the S&P 500 Index. The XLK has a 4-star in the Morningstar rating by March 2026, indicating an above-average performance that the XLK is in the top 22.5% to 32.5% of technology category. The XLK ETF provides precise exposure to companies from technology hardware, storage and peripherals; software; communications equipment; semiconductors and semiconductor equipment; IT services; and electronic equipment, instruments and components industries (State Street). This feature allows investors to take strategic or tactical positions at a more targeted level than traditional style based investing. The companies in the XLK are selected from the S&P 500 based on the Global Industry Classification Standard (GICS) where companies are assigned to a single classification based on revenue.

3.2 Data Collection

Using the dataset from the investing website, we derive the following variables for the analysis:

- `Price` (in USD): The final price at which the ETF traded when the stock exchange closed (typically 4:00 PM EST).
- `Open` (in USD): The price of the very first transaction when the market opened (9:30 AM EST).
- `High` & `Low` (in USD): The maximum and minimum prices reached during the trading session.
- `Vol.`: The total number of shares traded during the day (e.g., 15.90M means 15.9 million shares).
- `Change`: The percentage difference between the current day's closing price and the previous day's closing price.

The dataset comprises daily observations spanning from January 3, 2017, to March 28, 2026. This time frame was specifically selected to ensure that the launch of ChatGPT (November 30, 2022) serves as a central pivot point, providing a balanced sample size for comparing pre- and post-innovation market dynamics

3.3 Data Cleaning

In the raw dataset from the investing website, we observe the point information on weekends are carried over from the Friday of the same week with empty number in `Vol`. We removed these points since they are not consistent with the general pattern and can be misleading in identifying the long-term price fluctuation. We then split the dataset into two groups, using the time stamp November 30, 2022 (the release date of chatGPT) as the cutoff.

Table 1: Summary Statistics of Stock Price

Mean Price	Standard deviation	Minimum	Maximum
69.91683	33.95005	24.4	152.07

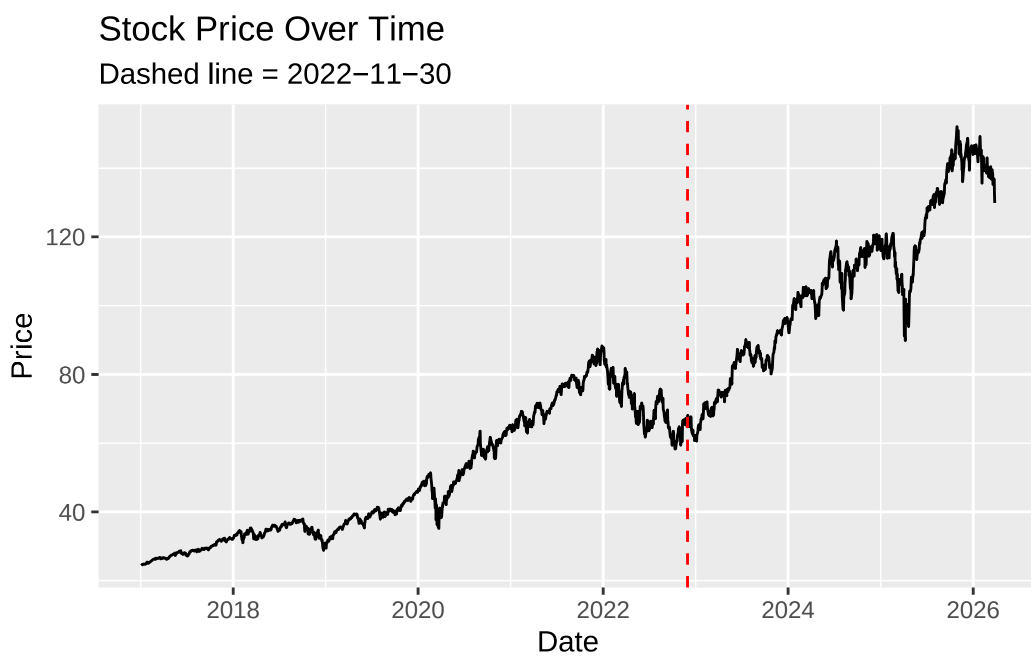

Figure 1: Raw Stock Price Over Time

Table 1 summarized the descriptive statistics for the stock price from January 3, 2017, to March 27, 2026, including the mean, standard deviation, and range. A more intuitive historical price trajectory is illustrated in Figure 1, where the vertical red dashed line marks the official release of ChatGPT. Overall, the stock shows a sustained upward trend throughout the observed period.

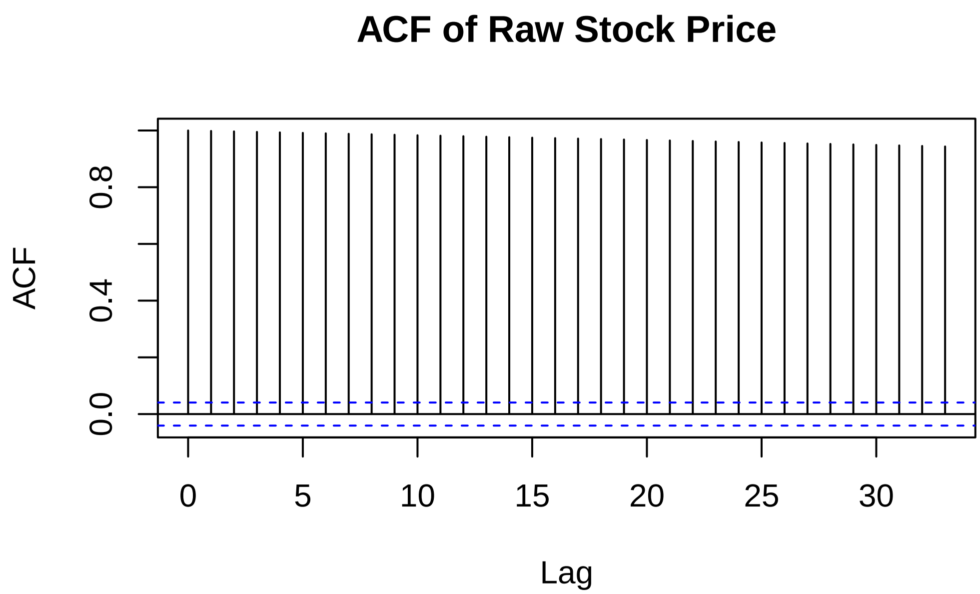

Figure 2: Autocorrelation Function (ACF) of Raw Stock Price.

The autocorrelation function (ACF) plot in Figure 2 demonstrates a strong correlation between data points, indicating this time series is non-stationary and does not exhibits white noise characteristics. The ACF values persist well beyond the significance boundaries and decay at a very slow rate. This high correlation is explainable since the stock prices are accumulated based on the previous day's close price. Therefore, data transformation is required to achieve stationarity prior to modeling.

4 Methods and Results

4.1 Fitting a regression model on price

We fit a regression model with a dummy variable D_t as the predictor.

$$\widehat{Price} = \hat{\beta}_0 + \hat{\beta}_1 t + \hat{\beta}_2 D_t + \hat{\beta}_3 t D_t$$

where $D_t = 0$ represents the period before the appearance of ChatGPT, and $D_t = 1$ represents the period after.

As shown in Figure 3, the two regression lines display different slopes across the two periods. The slope of price over time is steeper after the appearance of AI, meaning the price increased faster during this period.

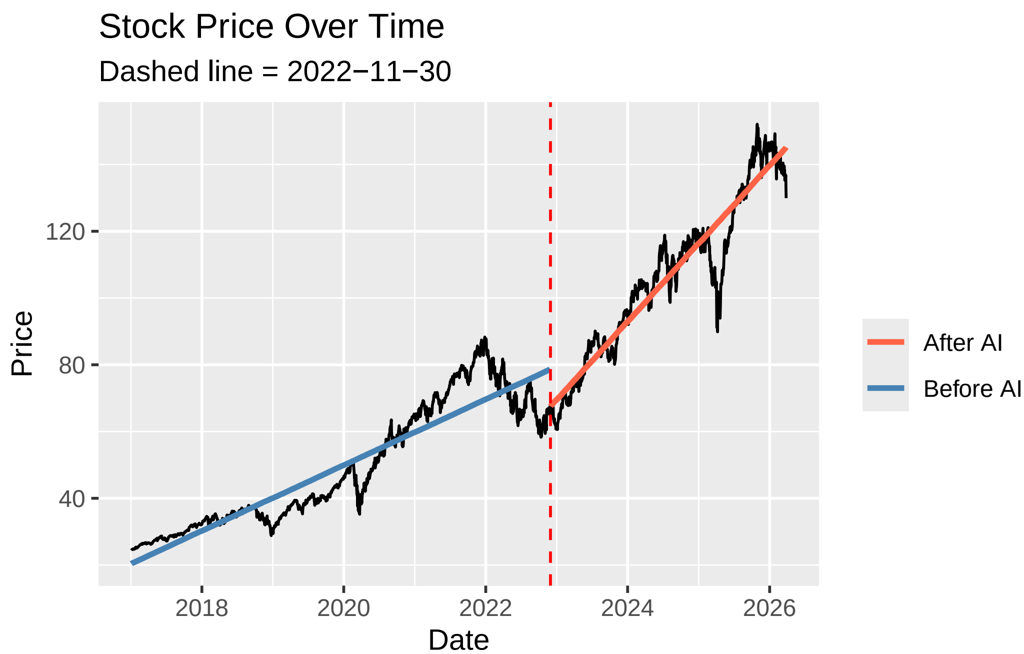

Figure 3: Fitted regression lines for price over time, separated by pre and post-AI periods.

However, the assumptions on regression were not satisfied. The price data is non-stationary, and the QQ plot of the residuals indicates a clear deviation from normality. The Augmented Dickey-Fuller (ADF) test on the model residuals yielded a p-value of 0.0575, failing to reject the null hypothesis of a unit root using 0.01 as the significance level. This suggests that the residuals remain non-stationary.

A robust regression approach was considered too. However, since robust methods do not resolve issues of non-stationarity, this approach was not considered in further analysis. As the residuals are not stationary, an ARMA model cannot be appropriately fitted, and thus no further modelling was performed with regression.

4.2 Differencing on log price

To stabalize the variance and ensure stationarity, we applied a natural logarithmic transformation. The first difference of the log price series, known as the log return, was then used to remove the trend and produce a stationary series suitable for model estimation. Figure 4 shows the log return over time. While the log return mostly fluctuates around a mean of zero with small fluctuations, there are few large spikes. The biggest fluctuations occur around 2020 when the pandemic began, suggesting that the pandemic can might casue substantial uncertainty and sharp movements in the market.

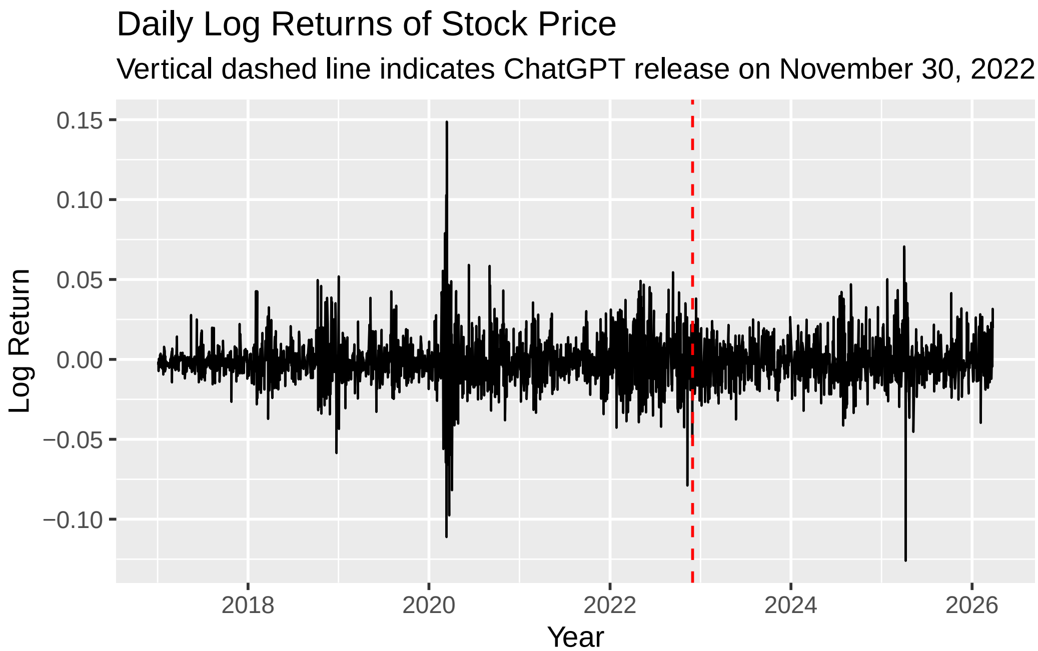

Figure 4: Daily Log Returns of Stock Price (2017–2026).

Although, in Figure 5, the ACF shows there still exists few autocorrelation coefficients outside the significance boundaries, the ADF test was conducted for a formal assessment. The test yielded a p-value of 0.01, which rejects the null hypothesis of a unit root and confirms that the log return series is stationary.

4.3 Using Pre-AI Prices to Predict Post-AI Prices

We aimed to examine whether the emergence of AI had an unexpected impact on the stock market and contributed to increased price fluctuations. To investigate this, we used stock price data from January 3, 2017 to November 30, 2022 to forecast stock prices after November 30, 2022.

First, we fitted ARIMA models using the historical price data from January 3, 2017 to November 30, 2022. We used AIC, AICc and BIC metrics to determine the best model for forecasting.

We then applied three “benchmark” forecasting methods: the mean method, the naive method, and the seasonal naive method. Mean method forecasts all future values as equal to the average

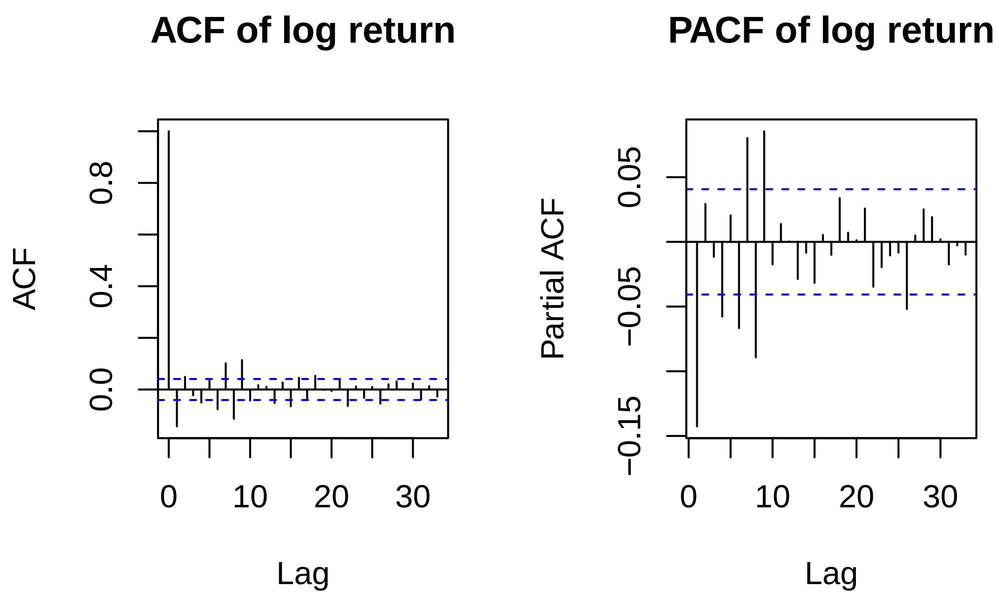

Figure 5: Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of Log Return.

of the training set. Naive method set all forecast values to be the value of the last observation of the historical data. Seasonal naive method set each forecast to be equal to the last observed value from the same season. In this case, we tested for weekly seasonality using this method.

Finally, we compared the performance of the ARIMA model and the benchmark methods using point comparison metrics like MAE, RMSE, MASE and MASSE.

Using the ACF and PACF plots, we can determine the appropriate order of the ARIMA model. According to Figure 5, the ACF gradually tails off, suggesting the presence of autoregressive behavior. According to the PACF, there are strong spikes at lags 2, 4, 7, and 8, and several other significant lags. Based on these patterns, ARIMA(1,1,1), ARIMA(2,1,2), ARIMA(4,1,4), ARIMA(6,1,6), ARIMA(7,1,7), and ARIMA(8,1,8) models were fitted to the log price data to capture the time series.

The p values for Ljung-Box Statistic plots indicate that ARIMA(1,1,1) and ARIMA(2,1,2) are not good fits, because p values are significant for all lags. ACF plots for ARIMA (4,1,4), ARIMA(7,1,7), and ARIMA (8,1,8) looks stationary, but QQ plots show the points are deviated from the reference line, meaning the normality condition is not met. To determine which model performs the best, we compare the model AIC, AICc, BIC, and log likelihood values.

Table 2: Comparison of Model Performance

	AIC	AICc	BIC	Log Likelihood
ARIMA(1,1,1)	-5.417434	-5.417429	-5.406733	4030.863
ARIMA(2,1,2)	-5.463286	-5.463268	-5.445450	4066.953
ARIMA(4,1,4)	-5.467580	-5.467515	-5.435475	4074.146
ARIMA(6,1,6)	-5.466568	-5.466425	-5.420193	4077.393
ARIMA(7,1,7)	-5.471382	-5.471190	-5.417873	4082.973
ARIMA(8,1,8)	-5.468679	-5.468430	-5.408036	4082.963

We also compared the MAE, RMSE, MASE, and MASSE for all ARIMA models. There was no significant difference between models. Based on the AICc values, ARIMA (7,1,7) has the lower AICc values. Therefore, ARIMA(7,1,7) was picked as the best ARIMA model for forecasting.

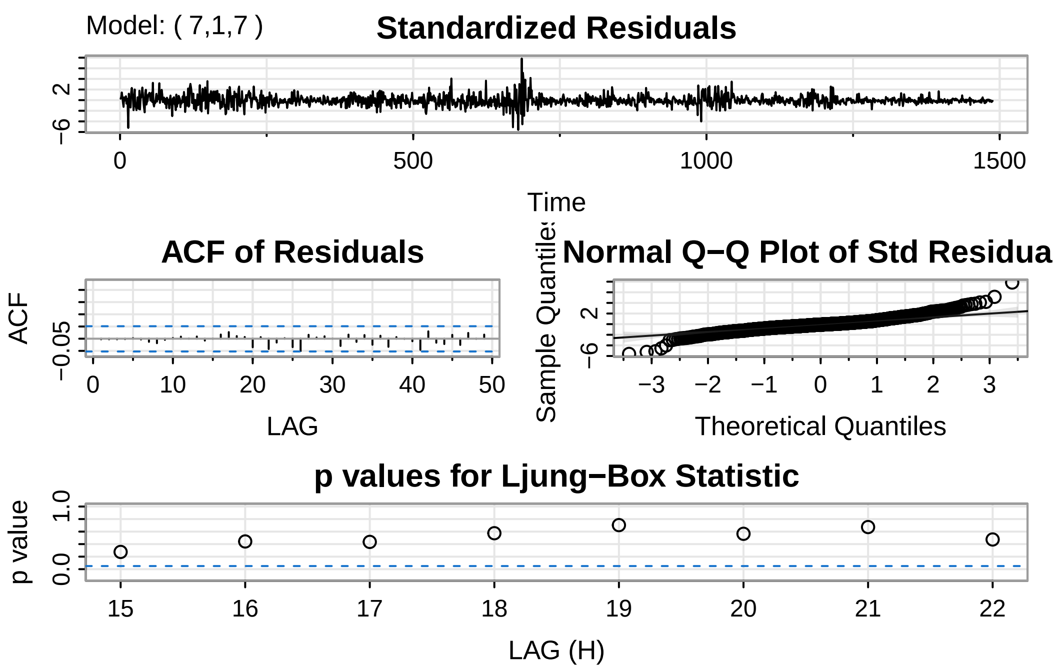

Figure 6: Diagnostic Plot for ARIMA(7,1,7)

Table 3: Forecast Errors

	MAE	RMSE	MASE	MASSE
Mean	0.01048	0.01456	0.64102	0.57560

Table 3: Forecast Errors

	MAE	RMSE	MASE	MASSE
Naive	0.01043	0.01457	0.63780	0.57599
Seasonal	0.02020	0.02639	1.23550	1.04291
ARIMA(7,1,7)	0.01049	0.01459	0.64166	0.57652

According to Table 3, MAE (Mean Absolute Error) and RMSE (Root Mean Squared Error), which measure the difference between actual and predicted values, are small for the mean method but also for naive and ARIMA models. Hence, a simple model like Mean method can effectively predict the data following the cutoff date using data prior to November 30, 2022. We assumed ARIMA would do a better job, because it's a more comprehensive model. But our results showed that simple model can be powerful and was good enough for this prediction.

Furthermore, this suggests that AI have not yet contribute a significant impact in the stock market, at least in terms of predictability using our proposed models. As a result, we will use the entire time frame, allocating 80% for training and 20% for testing, to build a forecast model for predicting prices in 2026.

4.4 Comparing Models Over the Full Time Frame

80% of data from January 3, 2017 to March 27, 2026 were used as the training set, and 20% of data were used for testing in building our models. Similar to section 3.3, ARIMA(1,1,1), ARIMA(2,1,2), ARIMA(4,1,4), ARIMA(7,1,7), and ARIMA(8,1,8) models were fitted. The best ARIMA model was determined using AIC, AICc, BIC, and log liklihood. In addition, three benchmark methods – the mean method, naive method, and seasonal naive method – were applied for comparison. The best-performing ARIMA model was then compared against these benchmark methods.

First, we fit the proposed ARIMA models. The diagnostics plots suggested that ARIMA(4,1,4), ARIMA(7,1,7) and ARIMA (8,1,8) might be the better model, because some p values of the Ljung box test were insignificant. To formally determine the most appropriate model structure, we compared the candidate models using the criteria shown in Table 4.

Table 4: Comparison of Model Performance

	AIC	AICc	BIC	logLik
ARIMA(1,1,1)	-5.477657	-5.477655	-5.470223	6357.082
ARIMA(2,1,2)	-5.492856	-5.492849	-5.480465	6376.713
ARIMA(4,1,4)	-5.503769	-5.503742	-5.481465	6393.372
ARIMA(7,1,7)	-5.503271	-5.503192	-5.466098	6398.794

Table 4: Comparison of Model Performance

	AIC	AICc	BIC	logLik
ARIMA(8,1,8)	-5.501308	-5.501206	-5.459179	6398.517

Among the candidate models, ARIMA(4,1,4) was selected as the best fitting model, as it achieved the lowest AIC, AICc, and BIC values. While ARIMA(7,1,7) produced a very similar AIC, its notably higher BIC indicates that the additional parameters are not justified, making ARIMA(4,1,4) the preferred specification.

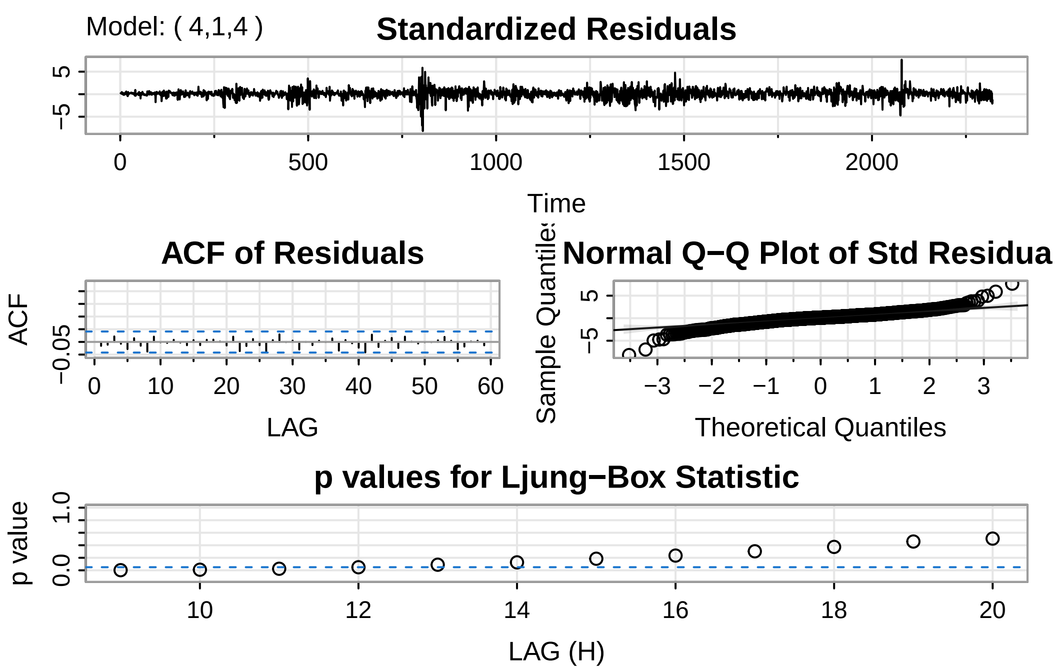

Figure 7: Diagnostic Plot for ARIMA(4,1,4)

Table 5: Forecast Errors - Full Model

	MAE	RMSE	MASE	MASSE
Mean	0.01127	0.01629	0.71762	0.68426
Naive	0.01119	0.01639	0.71208	0.68834
Seasonal	0.01350	0.01908	0.85921	0.80137
ARIMA(4,1,4)	0.01127	0.01629	0.71765	0.68418

Table 5 shows the naive method has the lowest MAE while the mean method has the lowest

RMSE. MAPE (Mean Absolute Percentage Error) is a percentage-based error metric that scales forecast errors relative to the actual values. However, MAPE(Mean Absolute Percentage Error) is not included in this comparison because the log return data fluctuates around zero, where some small values closer to zero make MAPE enormous and not a good parameter to look at. MASE (Mean Absolute Scaled Error) and MASSE (Root Mean Squared Scaled Error) are scaled error metrics, and they works by scaling errors based on the training MAE from a simple forecast method. This makes them stable because the denominator is a fixed constant derived from the training data, providing a more informative alternative when MAPE fails to give meaningful results. Based on the lowest MASE and RMSSE values, the mean method and ARIMA (4,1,4) performs best overall. Again, the simple models performed as good as ARIMA. The differences between models were extremely small. ARIMA model may not be needed to do forecasting in this case.

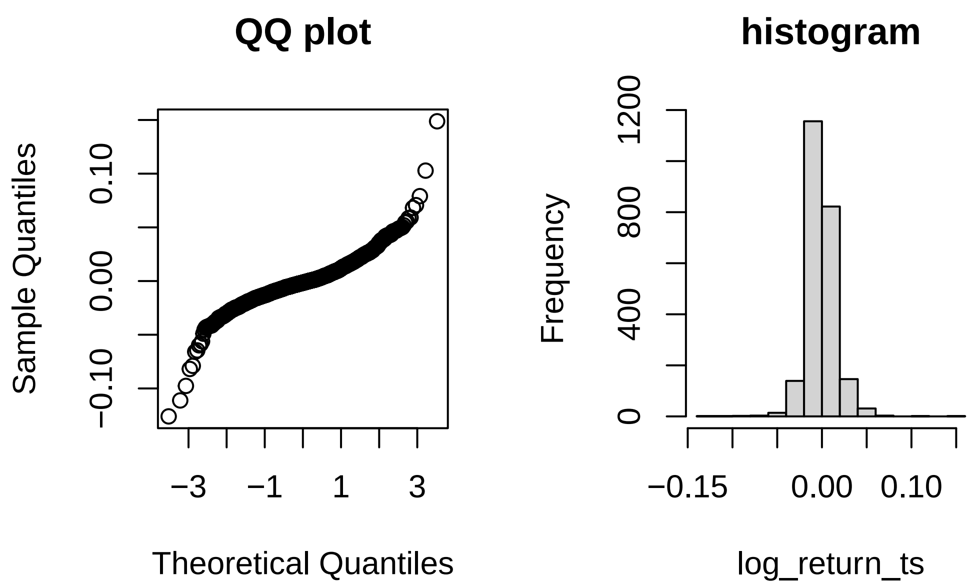

Figure 8: QQ plot and Histogram for log returns

Next, we construct a prediction interval for the mean method forecast model and ARIMA(4,1,4). In Figure 8, while the histogram suggests the residuals are approximately normally distributed, the QQ-plot indicates that the log returns deviate from normality, violating classic assumptions. Despite this, Figure 9 shows that the prediction interval successfully captures all observations in the test set, suggesting the interval is functioning as intended. However, the interval is extremely wide, reflecting high uncertainty in the forecasts. This indicates that while the mean method provides valid coverage, it offers little practical predictive power for long-term returns.

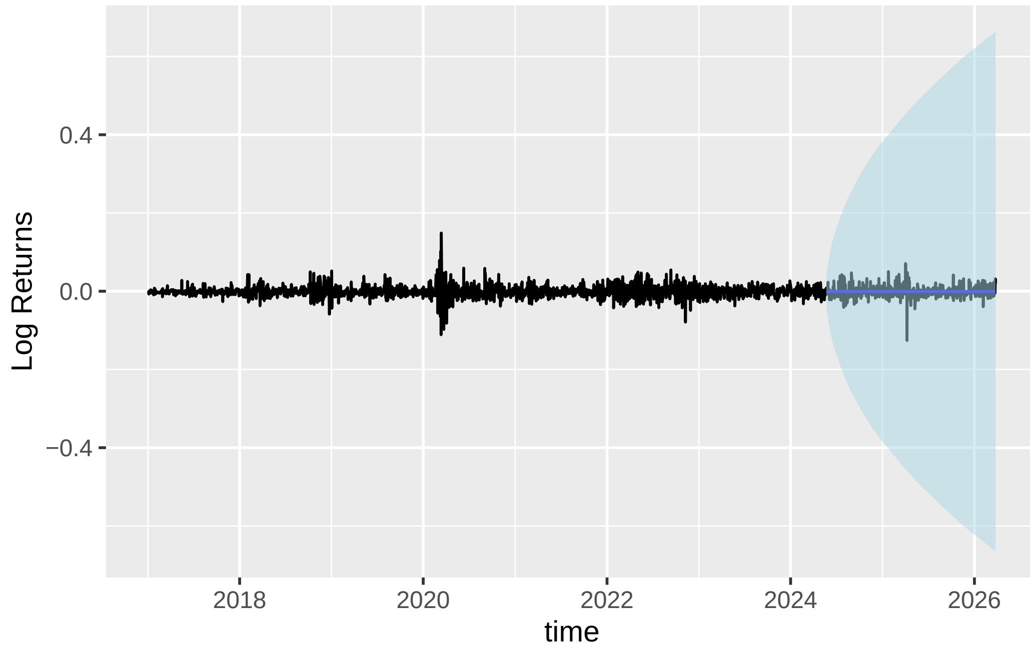

Figure 9: Mean method forecasts and prediction intervals for log returns

Figure 10 shows the forecast results using ARIMA (4,1,4) compared with the actual data. The forecast of ARIMA was close to zero. This was because the log returns had a mean close to zero. However, the prediction interval did not capture all the actual data, especially during period of extreme volatility. Therefore, while ARIMA is effective in predicting the average behavior, it struggles to predict sudden large change in the data.

4.5 Forecast Future 2026 Data

Using ARIMA(4,1,4), we forecast the stock price for the next 30 days starting March 28, 2026. The predicted values appeared relatively flat, which is expected for ARIMA models over short period of time. Since future white noise terms are unknown, the forecasts tend to settle toward a constant, resulting in a smoother and less stochastic pattern compared to the real data, as shown in Figure 11. Although this is a simple model, the forecasts are still useful for understanding the potential short-term behavior and overall trend of the market. Investors and analysts may use these forecasts as a reference for anticipating future movement patterns, and supporting short-term decision making.

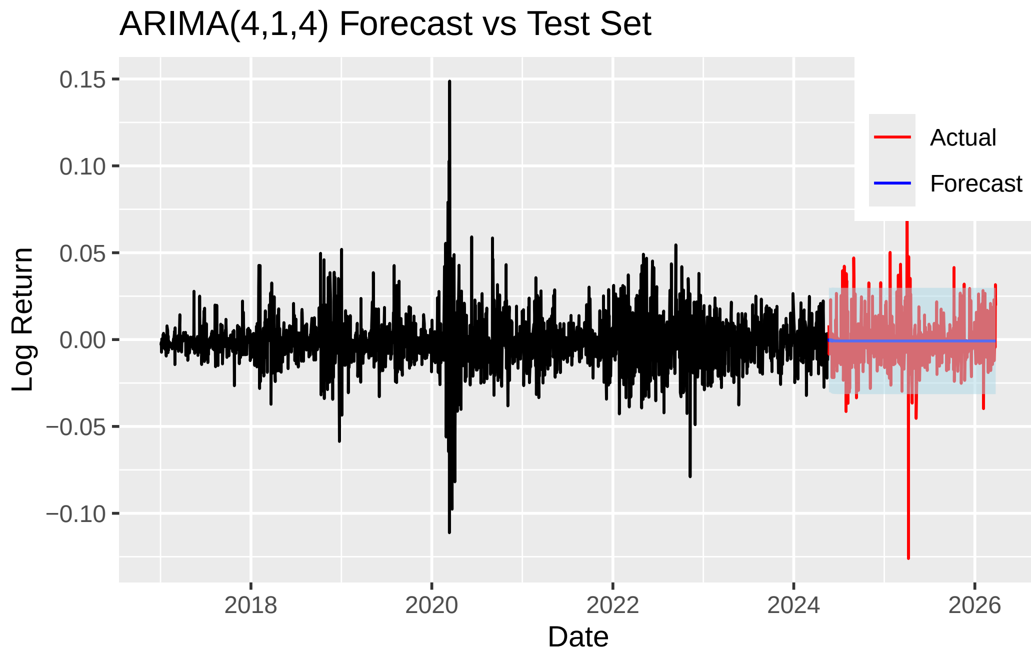

Figure 10: ARIMA(4,1,4) Forecast Data Compared with Actual Data

5 Conclusions

This project used stock price data from January 3, 2017 to March 27, 2026 to study stock price behavior and forecasting performance using ARIMA models and benchmark forecasting methods. The first part of the project investigated whether the emergence of AI, specifically following the launch of ChatGPT on November 30, 2022, had a meaningful impact on stock market. To examine this question, we used data before November 30, 2022 to forecast stock prices after this date. Several ARIMA models and three benchmark forecasting methods were proposed and evaluated. The results showed that the performance of the best performing ARIMA models was similar to that of the simpler benchmark methods. Overall, the models were able to forecast the post-AI period reasonably well using pre-AI data, suggesting that the introduction of AI did not produce a significant change in the stock price. Therefore, we used the full data frame for forecasting.

In the second part, we used the full time frame from January 3, 2017 to March 27, 2026 to further evaluate forecasting performance. Multiple ARIMA models and benchmark methods were compared accuracy metrics. The results again indicated no substantial difference in predictive performance among the models. Among the ARIMA models, ARIMA(4,1,4) was selected as the preferred model based on model selection criteria, and it was used to forecast stock prices for the following 30 days. The forecasting results were shown for understanding the potential short-term behavior.

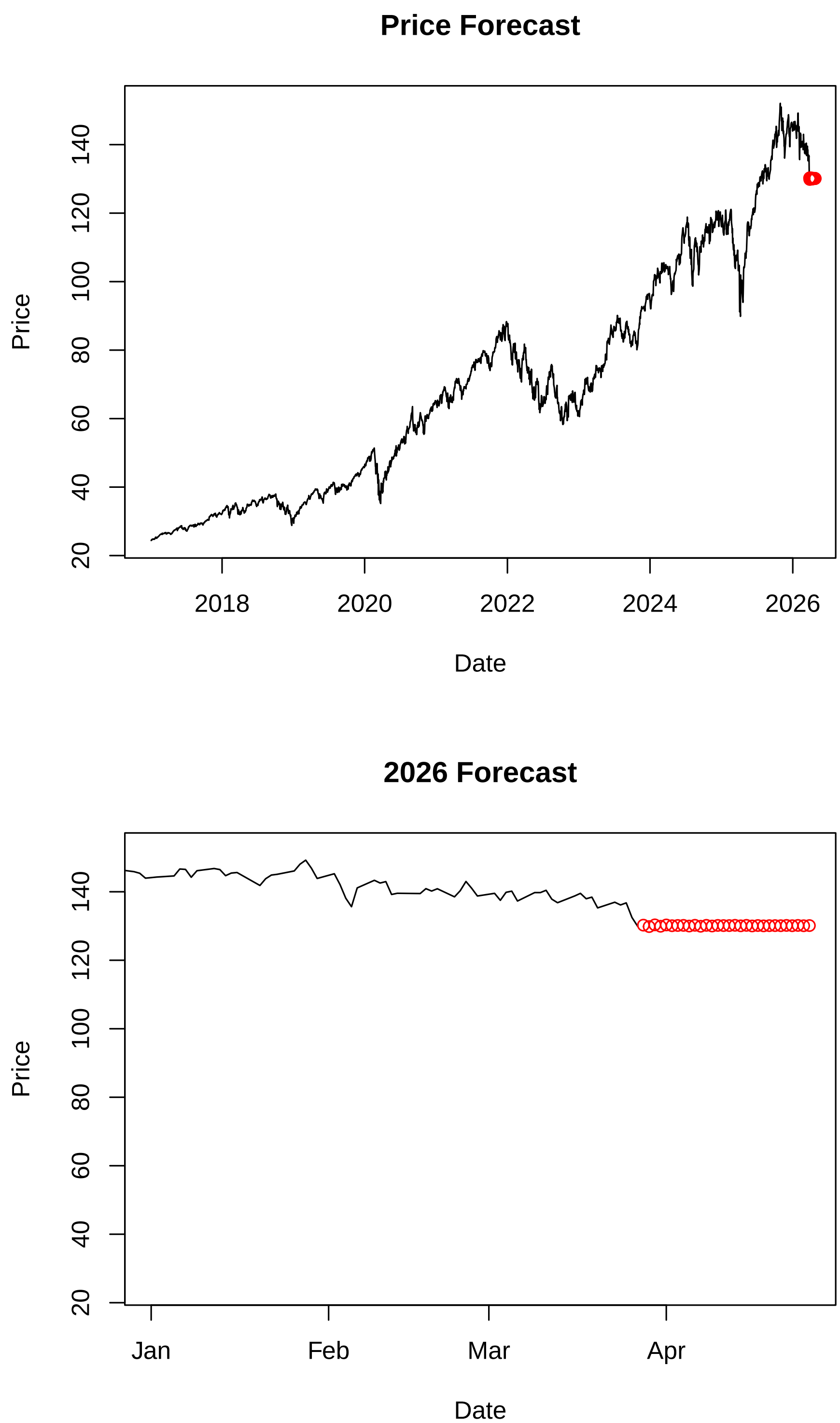

Figure 11: ARIMA(4,1,4) fitted values and forecasts from 2017 to 2026, with a closer view of the 2026 forecast period

6 Limitations

This study has several limitations. First, ARIMA models are designed to capture patterns within the historical structure of a time series, but technological breakthroughs such as generative AI may create nonlinear or structural changes that are difficult to detect using traditional forecasting models. Therefore, the finding that ARIMA models performed similarly to benchmark methods does not necessarily imply that AI had no impact on the stock market. Instead, the effects of AI may appear through changes in long-term valuation, investor expectations, or sector-wide capital allocation rather than short-term return predictability.

Second, XLK represents a broad technology-sector ETF containing firms with very different levels of AI exposure. While some companies, such as NVIDIA and Microsoft, experienced substantial market revaluation related to AI development, other firms within the ETF may have been less affected. Aggregating all firms into a single sector-level index may dilute firm-specific responses to AI and reduce the visibility of technological shocks in the overall time series.

Another limitation is the use of November 30, 2022, the release date of ChatGPT, as the single cutoff point for identifying a potential structural break. Financial markets may react gradually to technological innovation, and investors may have partially priced in AI expectations before the public release of ChatGPT. In addition, later developments such as GPT-4, AI infrastructure expansion, and semiconductor demand may also contribute to changes in technology-sector valuation. Therefore, AI adoption may be better understood as an evolving process rather than a single event.

In addition, the post-AI observation period remains relatively short compared to the long-term nature of technological revolutions discussed in prior literature. Historical innovations such as the microprocessor revolution and the information technology revolution unfolded over many years. As a result, the long-term effects of generative AI on market structure and productivity may not yet be fully observable within the current study period.

Finally, stock market returns are inherently noisy and difficult to predict. Financial return series often behave similarly to random walks, meaning even substantial economic or technological changes may not produce large improvements in forecasting accuracy. Consequently, the similarity in performance between ARIMA and benchmark methods may partly reflect the fundamental unpredictability of short-term stock returns rather than the absence of AI-related market effects.

7 Future Directions

To account for external factors that might affect the stock market, we could add these as predictors in our model. This can help us better estimate stock prices.

Although we have not been able to conclude AI has significant impact on the behavior of technology sector stock returns, it is still useful to estimate how sensitive technology sector stock price comparing to the broader market indices or other sectors. We might expect the technology sector to show higher volatility and respond more to growth expectations. In future studies, we can fit and compare separate models for different market segments, such as the S&P 500 and individual sectors like healthcare or financials, to better understand these differences.

It is also possible to incorporate the method proposed by Campos-Martins (2024) with our model. After identifying notable development by AI companies, we can adjust our predictors to improve accuracy. For example, we could include an indicator variable that captures major AI events or achievements. This would allow the model to account for changes over time driven by new innovations, instead of relying only on fixed predictors.

Finally, our research should follow a consistent process where we constantly refit the models as the stock market changes over time. Updating the model with new data can help us see whether there are any long-term effects or clear shifts in the data, especially as AI continues to develop and become more widely used.

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